

Paul Jerabek

Group Leader Computational Materials Design
Deputy Head of Department of Materials Design

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Education

- Dec 2010 – Sep 2014 **Dr. rer. nat.**, *Philipps-Universität Marburg*
Thesis: *Clusterartige und hochkoordinierte Metallkomplexe*
- Jun 2010 – Nov 2010 **Dipl.-Chem.**, *Philipps-Universität Marburg*
Thesis: *Bindungstheoretische Untersuchungen der Systeme des Typs YE-X-EY*
- Oct 2005 – Jun 2010 **Studies of Chemistry**, *Philipps-Universität Marburg*

Professional experience

- Sep 2022 – present **Deputy Head of Department of Materials Design**, *Helmholtz-Zentrum Hereon*
Department administration and representation, third-party funding acquisition, research strategy planning, digital transformation.
- Oct 2021 – present **Co-Leader, German-New Zealand Green Hydrogen Centre (He Honoka Hauwai)**, *Helmholtz-Zentrum Hereon / University of Otago*
Administration and representation, strategic planning, networking, funding acquisition, event organization.
- Sep 2019 – present **Group Leader “Metal Hydride Modeling”**, *Helmholtz-Zentrum Hereon*
First-principles molecular dynamics, CALPHAD and phase-field simulations for the characterization and optimization of novel metal hydride materials for large-scale hydrogen storage. Close collaborations with Lawrence Livermore National Laboratory (USA) and the University of Otago (New Zealand).
- Jul 2018 – Aug 2019 **Humboldt Return Fellow**, *Max-Planck-Institut für Kohlenforschung*
Theoretical investigation of physical properties of organonitrogen; substituent and solvent effects on switching temperature in Fe(II) spin-crossover complexes; reaction mechanism exploration for asymmetrical, homogeneous catalysis; bonding situation and electronic properties of pincer complexes.
- Dec 2017 – Jun 2018 **Postdoctoral Research Assistant**, *Massey University*
Monte-Carlo melting simulations of heavy and superheavy elements; first-principles molecular dynamics simulations of coinage metals in liquid and solid phase; highly accurate calculations of atomic polarizabilities.
- Dec 2015 – Nov 2017 **Feodor Lynen Research Fellow**, *Massey University*
Photochemistry of small molecules in different environments; influence of relativistic and solvent effects on atomic and molecular bonding properties; development of highly accurate ab-initio many-body potentials for heavy and superheavy elements.
- Dec 2014 – Nov 2015 **Postdoctoral Research Assistant**, *Philipps-Universität Marburg*
Method development (C) for core electron spectroscopy of organic molecules on surfaces; density functional theory based investigation of transition metal clusters and complexes.

Dec 2010 – Nov 2014	Research Assistant, <i>Philipps-Universität Marburg</i> Quantum chemical investigation of main-group and transition metal compounds in collaboration with experimental groups in Bochum, München, Göttingen, Dresden, Münster and San Diego. Redesign of the lab course in Theoretical Chemistry.
Jan 2007 – Dec 2010	Student Assistant, <i>Philipps-Universität Marburg</i> Supervision of lab courses in Physical Chemistry and Chemistry for Medical Students.

Third-party funding

2024	Sustainable TiFe Generation from Secondary Sources BMBF (Germany) Principal Investigator 1.5 M€ 3 years
2024	HIDA Visiting Researcher Grant (with S. Faouri) Helmholtz Association Host 10 k€ 2 months
2023	HIDA Visiting Researcher Grant (with A. Rowberg) Helmholtz Association Host 15 k€ 3 months
2022	Generating Hydrogen Storage Materials from New Zealand Resources BMBF (Germany) Coordinator 400 k€ 3 years
2021	Establishing a German-New Zealand Research Centre for Green Hydrogen BMBF (Germany) Coordinator 760 k€ 5 years
2021	HIDA Visiting Scientist Grant (with M. Selzer) Helmholtz Association Fellow 10 k€ 3 months
2017	Humboldt Return Fellowship Alexander von Humboldt Foundation Fellow 47 k€ 1 year
2016	Computation of highly accurate interatomic potentials of heavy and superheavy elements New Zealand eScience Infrastructure (NeSI) Principal Investigator 117 k€ 1.5 years
2014	Hochpräzise quantenchemische Simulationen von atomaren und molekularen Phasen unter extremen Bedingungen Alexander von Humboldt Foundation Fellow 85 k€ 2 years

Conference organization

2025	E-MRS Fall Meeting 2025 — Symposium A Co-organizer, Warsaw, Poland
2024	E-MRS Fall Meeting 2024 — Symposium C: Sustainable materials for chemical and electrochemical energy storage II Main organizer, Warsaw, Poland
2023	Inaugural New Zealand Hydrogen Symposium Co-organizer, Dunedin, New Zealand
2015	Bonding Analysis Workshop Conceptualization and main organizer, Marburg, Germany

Teaching

2025 SoSe	Hydrogen Technology Co-lecturer, Technische Universität Hamburg (TUHH), Hamburg
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2023 – present	Lectureship Lecturer, Technische Hochschule Mittelhessen, Gießen
2019 – 2019	Knowledge for Tomorrow – Cooperative Research Projects in Sub-Saharan Africa (Volkswagen-Stiftung) Tutor, University of Mauritius, Reduit
2014 – 2019	Hands-on workshops on electronic structure methods Workshop lecturer, Various universities, Hyderabad, Dunedin, Denton, Mariapfarr, Mauritius
2010 – 2014	Practical course in Theoretical Chemistry Teaching assistant, Philipps-Universität Marburg, Marburg
2007 – 2010	Physical Chemistry and Chemistry for Medical Students Student assistant, Philipps-Universität Marburg, Marburg

Talks and posters

May 2025	German-New Zealand WTZ Meeting 2025 , Wellington, New Zealand Invited Talk
Feb 2025	11th International Conference on Advanced Materials and Nanotechnology (AMN11) , Christchurch, New Zealand Invited Talk
Jul 2024	New Zealand-Germany Energy Dialogue 2024 , Berlin, Wellington, Germany, New Zealand Invited Talk
Feb 2023	10th International Conference on Advanced Materials and Nanotechnology (AMN10) , Rotorua, New Zealand Invited Talk
Jul 2018	18th International Conference on Computational and Mathematical Methods in Science and Engineering (CMMSE-2018) , Rota, Spain Invited Talk
Jul 2016	16th International Conference on Computational and Mathematical Methods in Science and Engineering (CMMSE-2016) , Rota, Spain Invited Talk
Mar 2024	German Physical Society (DPG) Spring Meeting 2024 , Berlin, Germany Contributed Talk
Jan 2024	2nd New Zealand Hydrogen Symposium (NZHS-2) , Wellington, New Zealand Contributed Talk
Mar 2023	German Physical Society (DPG) Spring Meeting 2023 , Dresden, Germany Contributed Talk
Sep 2022	European Materials Research Society (E-MRS) Fall Meeting 2022 , Warsaw, Poland Contributed Talk
Mar 2019	Chemiedozententagung 2019 , Koblenz, Germany Contributed Talk
Nov 2017	17th Biennial Symposium of the Australian and New Zealand Associations of von Humboldt Fellows , Wellington, New Zealand Contributed Talk
Sep 2017	12th International Conference on Relativistic Effects in Heavy-Element Chemistry and Physics (REHE) , Marburg, Germany Contributed Talk

Aug 2016	New Zealand Institute of Chemistry Conference (NZIC-16) , Queenstown, New Zealand Contributed Talk
May 2024	18th International Symposium of Metal-Hydrogen Systems (MH 2024) , Saint-Malo, France Poster
Sep 2018	54th Symposium on Theoretical Chemistry (STC) , Halle (Saale), Germany Poster
Aug 2017	11th World Association of Theoretical and Computational Chemists (WATOC) , Munich, Germany Poster
Jul 2016	9th Congress of the International Society for Theoretical Chemical Physics (ISTCP IX) , Grand Forks (ND), USA Poster
Sep 2015	51st Symposium on Theoretical Chemistry (STC) , Postdam, Germany Poster
Sep 2013	49th Symposium on Theoretical Chemistry (STC) , Erlangen, Germany Poster
Sep 2012	48th Symposium on Theoretical Chemistry (STC) , Karlsruhe, Germany Poster
Oct 2011	Modeling of Molecular Properties Conference (MolMod) , Heidelberg, Germany Poster
Jul 2011	9th World Association of Theoretical and Computational Chemists (WATOC) , Santiago de Compostela, Spain Poster
Sep 2010	46th Symposium on Theoretical Chemistry (STC) , Münster, Germany Poster
Feb 2025	University of Waikato , Hamilton, New Zealand Invited Seminar
Jun 2024	University of Hamburg , Hamburg, Germany Invited Seminar
Feb 2024	Massey University , Auckland, New Zealand Invited Seminar
Feb 2024	University of Waikato , Hamilton, New Zealand Invited Seminar
Mar 2023	Massey University , Auckland, New Zealand Invited Seminar
Mar 2023	University of Auckland , Auckland, New Zealand Invited Seminar
Dec 2021	Karlsruhe Institute of Technology (KIT) , Karlsruhe, Germany Invited Seminar
May 2019	Helmholtz-Zentrum für Material- und Küstenforschung , Geesthacht, Germany Invited Seminar
Apr 2019	Universität Graz , Graz, Austria Invited Seminar
Feb 2019	Technische Universität Berlin , Berlin, Germany Invited Seminar

Jan 2019	GSI Helmholtzzentrum für Schwerionenforschung , Darmstadt, Germany Invited Seminar
Feb 2018	University of Otago , Dunedin, New Zealand Invited Seminar
Jun 2017	University of Otago , Dunedin, New Zealand Invited Seminar
Jul 2016	University of North Texas , Denton (TX), USA Invited Seminar
Oct 2019	Summer School/Workshop on Computational Chemistry “Knowledge for Tomorrow – Cooperative Research Projects in Sub-Saharan Africa” , Reduit, Mauritius Workshop Lecture
Mar 2019	XIIth Workshop on Modern Methods in Quantum Chemistry , Mariapfarr, Austria Workshop Lecture
Dec 2017	University of Otago , Dunedin, New Zealand Workshop Lecture
Jul 2016	University of North Texas , Denton (TX), USA Workshop Lecture
Jan 2014	MCBR Winter School , Hyderabad, India Workshop Lecture

Peer-reviewed journal articles

70. “Evaluation of the synergetic effects of Ni and Cr in $\text{TiFe}_{(1-x-y)}\text{Cr}_x\text{Ni}_y$ for enhanced hydriding thermodynamics”: E. Pericoli, V. Ferretti, E. Alvares, P. Jerabek, C. Pistidda, L. Pasquini, *Adv. Energy Sustain. Res.*, **7**, e70243, 2026.
69. “Relativistic effects are crucial for uraniumphosphorus bonding - a DFT study of cyclopentadienyluranium-phosphine complexes”: J.-E. Ahrens, P. Jerabek, L. Vondung, *Mol. Phys.*, e2611400, 2026.
68. “Role of Elemental Substitution on Hydrogen Incorporation in Characteristic Ti-Fe-Based Oxides”: A. J. E. Rowberg, S. Kang, K. Sellschopp, P. Jerabek, T. W. Heo, B. C. Wood, *ACS Appl. Energy Mater.*, **9**, 1242–1253, 2026.
67. “Leveraging atomic disorder to modulate hydrogen storage thermodynamics in compositionally complex intermetallics”: Y. Shang, A. Santhosh, P. Jerabek, T. Klassen, C. Pistidda, *Interdiscip. Mater.*, **0**, 1-13, 2026.
66. “Predicting hydrogen storage capacity loss of Si-containing FeTi alloys through ab initio thermodynamic modeling”: L. Dinachandran, E. Alvares, K. Sellschopp, T. Klassen, P. Jerabek, A. L. Garden, *Acta Mater.*, **301**, 121514, 2025.
65. “TiAlNb alloy interatomic potentials - comparing passive and active machine learning techniques with MTP and DeepMD”: A. Chandran, A. Santhosh, P. Jerabek, R. C. Aydin, C. J. Cyron, *Front. Mater.*, **12**, 2025.
64. “Low-temperature regeneration of metal hydride after impurity gas exposure”: Y. Shang, A. Santhosh, M. Ante, L. Wimbert, P. Jerabek, T. Schupp, T. Klassen, C. Pistidda, *Acta Mater.*, **298**, 121398, 2025.
63. “High Ionic Conduction in Rb- and Cs-Mixed Cation Amide for Energy Storage”: T. T. Le, K. Sellschopp, F. Murgia, A. L. Garden, S. Bordignon, J. P. Embs, M. R. Chierotti, A. Schökel, F. Karimi, P. Jerabek, T. Klassen, C. Pistidda, *Small*, 2502943, 2025.

62. "A Thermodynamic Approach to Model Multicomponent FeTi-based Alloys for Hydrogen Storage": E. Alvares, A. J. E. Rowberg, K. Sellschopp, B. C. Wood, T. Klassen, P. Jerabek, C. Pistidda, *Scr. Mater.*, **259**, 116516, 2025.
61. "Multiscale Modeling the FeTi Metal-Hydride Interphase Boundaries – Quantification of Decoupled Chemo-Mechanical Energies": E. Alvares, K. Sellschopp, B. Wang, S. Kang, T. Klassen, T. W. Heo, T. Klassen, P. Jerabek, C. Pistidda, *npj Comput. Mater.*, **10**, 249, 2024.
60. "Comparative analysis of ternary TiAlNb interatomic potentials - moment tensor vs. deep learning approaches": A. Chandran, A. Santhosh, C. Pistidda, P. Jerabek, R. Aydin, C. J. Cyron, *Front. Mater.*, **11**, 1466793, 2024.
59. "Thermodynamic assessment of the Ce-H and CeNi₅-H system": P. Hannappel, E. Alvares, F. Heubner, C. Pistidda, P. Jerabek, T. Weißgärber, *Calphad*, **85**, 102701, 2024.
58. "Predictable Electronic Tuning of FeII versus RuII Complexes via Choice of Azine Donor - An Electrochemical and Theoretical Study": M. Robb, L. Bondi, S. Rodriguez-Jimenez, A. Garden, P. Jerabek, S. Brooker, *Dalton Trans.*, **53**, 1999-2007, 2024.
57. "Tipping the balance between the bcc and fcc phase within the alkali and coinage metal groups": A. Robles-Navarro, P. Jerabek, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **63**, e202313679, 2024.
56. "First-principles study on interfacial property in MgB₂-based reactive hydride composites": Y. Shang, A. Santhosh, P. Jerabek, T. Klassen, C. Pistidda, *Scr. Mater.*, **240**, 115837, 2024.
55. "Low-Cost, Multifunctional, and Sustainable Sodium Sulfido Ferrate(II)": M. Ghazanfari, K. Siemensmeyer, A. Santhosh, J. Vrijmoed, M. Tallu, S. Dehnen, P. Jerabek, G. Thiele, *Inorg. Chem.*, **62**, 15358-15366, 2023.
54. "Ultra-lightweight compositionally complex alloys with large ambient-temperature hydrogen storage capacity": Y. Shang, Z. Lei, E. Alvares, S. Garroni, R. Dore, M. Rustici, S. Enzo, A. Schökel, Y. Shi, P. Jerabek, Z. Lu, T. Klassen, C. Pistidda, *Mater. Today*, **67**, 113-126, 2023.
53. "Influence of near-surface oxide layers on TiFe hydrogenation - Mechanistic insights and implications for hydrogen storage applications": A. Santhosh, S. Kang, N. Keilbart, B. C. Wood, T. Klassen, P. Jerabek, M. Dornheim, *J. Mater. Chem. A*, **11**, 18776-18789, 2023.
52. "Experimental and Computational Studies on the Formation of Mixed Amide-Hydride Solid Solutions for CsNH₂-CsH System": T.-T. Le, A. Santhosh, S. Bordignon, M. R. Chierotti, P. Jerabek, T. Klassen, C. Pistidda, *Results in Engineering*, **17**, 100895, 2023.
51. "Magnesium- and intermetallic alloys-based hydrides for energy storage: modelling, synthesis and properties": L. Pasquini, K. Sakaki, E. Akiba, P. Jerabek et al., *Prog. Energy*, **4**, 32007, 2022.
50. "Relativistic Effects Stabilize Unusual Gold(II) Sulfate Structure via Auophilic Interactions": P. Jerabek, A. Santhosh, P. Schwerdtfeger, *Inorg. Chem.*, **61**, 13077-13084, 2022.
49. "Solving a problem with a single parameter: A smooth bcc to fcc phase transition for metallic lithium": P. Jerabek, A. Burrows, P. Schwerdtfeger, *Chem. Commun.*, **58**, 13369-13372, 2022.
48. "Remarkable Infrared falsenlinear Optical, Dielectric, and Strong Diamagnetic Characteristics of Semiconducting K₃[BiS₃]": M. R. Ghazanfari, L. Vittadello, D. Al-Sabbagh, A. Santhosh, C. Frankcom, F. Fuß, C. A. von Randow, K. Siemensmeyer, J. C. Vrijmoed, F. Emmerling, P. Jerabek, M. Imlau, G. Thiele, *J. Phys. Chem. Lett.*, **13**, 6987-6993, 2022.

47. "Large-scale synthesis of mixed valance K₃[Fe₂S₄] with high dielectric and ferrimagnetic characteristics": M. R. Ghazanfari, A. Santhosh, J. C. Vrijmoed, K. Siemensmeyer, B. Peters, S. Dehnen, P. Jerabek, G. Thiele, *RSC Adv.*, **12**, 30514–30521, 2022.
46. "Large Exchange Bias, High Dielectric Constant, and Outstanding Ionic Conductivity in a Single-Phase Spin Glass": M. R. Ghazanfari, A. Santhosh, K. Siemensmeyer, F. Fuß, L. Staab, J. C. Vrijmoed, B. Peters, M. Liesegang, S. Dehnen, O. Oeckler, P. Jerabek, G. Thiele, *Adv. Electron. Mater.*, **8**, 2200483, 2022.
45. "From the gas phase to the solid state: The chemical bonding in the superheavy element flerovium": E. Florez, O. R. Smits, J.-M. Mewes, P. Jerabek, P. Schwerdtfeger, *J. Chem. Phys.*, **157**, 64304, 2022.
44. "An Effective Activation Method for Industrially Produced TiFeMn Powder for Hydrogen Storage": D. M. Dreistadt, L. Thi-Thu, G. Capurso, J. M. Bellosta von Colbe, A. Santhosh, C. Pistidda, N. Scharnagl, H. Ovri, C. Milanese, P. Jerabek, T. Klassen, J. Jepsen, *J. Alloys Compd.*, **919**, 165847, 2022.
43. "Quantitative assessment of ligand substituent effects on σ - and π -contributions to Fe-N bonds in spin crossover FeII complexes": L. Bondi, A. L. Garden, F. Totti, P. Jerabek, S. Brooker, *Chem. Eur. J.*, **28**, e202104314, 2022.
42. "Modeling the thermodynamics of the FeTi hydrogenation under para-equilibrium: An ab-initio and experimental study": E. Alvares, P. Jerabek, Y. Shang, A. Santhosh, C. Pistidda, T. W. Heo, B. Sundman, M. Dornheim, *Calphad*, **77**, 102426, 2022.
41. "Hydrogenation via a low energy mechafalsechemical approach: The MgB₂ case": C. Pistidda, A. Santhosh, P. Jerabek, Y. Shang, A. Girella, C. Milanese, M. Dore, S. Garroni, S. Bordigfalsen, M. R. Chierotti, T. Klassen, M. Dornheim, *J. Phys. Energy*, **3**, 44001, 2021.
40. "Oganesson: A Noble Gas Element That Is Neither Noble Nor a Gas": O. R. Smits, J. Mewes, P. Jerabek, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **59**, 23636–23640, 2020.
39. "First-principles melting of krypton and xenon based on many-body relativistic coupled-cluster interaction potentials": O. R. Smits, P. Jerabek, E. Pahl, P. Schwerdtfeger, *Phys. Rev. B*, **101**, 104103, 2020.
38. "Catalytic Asymmetric Fluorination of Copper Carbene Complexes: Preparative Advances and a Mechanistic Rationale": M. Buchsteiner, L. Martinez-Rodriguez, P. Jerabek, I. Pozo, M. Patzer, N. Nöthling, C. W. Lehmann, A. Fürstner, *Chem. Eur. J.*, **26**, 2509–2515, 2020.
37. "Quantitative and Chemically Intuitive Evaluation of the Nature of M–L Bonds in Paramagnetic Compounds: Application of EDA-NOCV Theory to Spin Crossover Complexes": L. Bondi, A. L. Garden, P. Jerabek, F. Totti, S. Brooker, *Chem. Eur. J.*, **26**, 13677–13685, 2020.
36. "Ligands Based on Phosphine-Stabilized Aluminum(I), Boron(I), and Carbon(0)": L. Vondung, P. Jerabek, R. Langer, *Chem. Eur. J.*, **25**, 3068–3076, 2019.
35. "Oganesson Is a Semiconductor: On the Relativistic Band-Gap Narrowing in the Heaviest falseble-Gas Solids": J.-M. Mewes, P. Jerabek, O. R. Smits, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **58**, 14260–14264, 2019.
34. "Solid Oganesson via a Many-Body Interaction Expansion Based on Relativistic Coupled-Cluster Theory and from Plane-Wave Relativistic Density Functional Theory": P. Jerabek, O. R. Smits, J.-M. Mewes, K. A. Peterson, P. Schwerdtfeger, *J. Phys. Chem. A*, **123**, 4201–4211, 2019.
33. "Dative and Electron-Sharing Bonding in Transition Metal Compounds": P. Jerabek, P. Schwerdtfeger, G. Frenking, *J. Comput. Chem.*, **40**, 247–264, 2019.

32. "A Hundred-Year-Old Experiment Re-evaluated: Accurate Ab-Initio Monte-Carlo Simulations of the Melting of Radon": O. Smits, P. Jerabek, E. Pahl, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **57**, 9961–9964, 2018.
31. "The Light-Driven Isomerization of Aqueous Nitrate: A Theoretical Perspective": J.-M. Mewes, P. Jerabek, D. S. Bohle, P. Schwerdtfeger, *ChemPhotoChem*, **2**, 725–733, 2018.
30. "Tipping the Balance between Ligand and Metal Protonation due to Relativistic Effects: Unusually High Proton Affinity in Gold(I) Pincer Complexes": P. Jerabek, L. Vondung, P. Schwerdtfeger, *Chem. Eur. J.*, **24**, 6047–6051, 2018.
29. "A relativistic coupled-cluster interaction potential and rovibrational constants for the xenon dimer": P. Jerabek, O. Smits, E. Pahl, P. Schwerdtfeger, *Mol. Phys.*, **116**, 1–8, 2018.
28. "Static dipole polarizability of palladium from relativistic coupled-cluster theory": P. Jerabek, P. Schwerdtfeger, J. K. Nagle, *Phys. Rev. A*, **98**, 12508, 2018.
27. "Electron and nucleon localization functions of oganesson: Approaching the Thomas-Fermi limit": P. Jerabek, B. Schuettrumpf, P. Schwerdtfeger, W. Nazarewicz, *Phys. Rev. Lett.*, **120**, 53001, 2018.
26. "Suppressed Phosphine Dissociation by Polarization Effects on the Donor–Acceptor Bonds in $[\text{Ni}(\text{PEt}_3)_{4-n}(\text{ECp}^*)_n]$ ($\text{E} = \text{Al}, \text{Ga}$)": J. Hornung, J. Weßing, P. Jerabek, C. Gemel, A. Pöthig, G. Frenking, R. A. Fischer, *Inorg. Chem.*, **57**, 12657–12664, 2018.
25. "Influence of Relativistic Effects on Bonding Modes in M(II) Dinuclear Complexes ($\text{M} = \text{Au}, \text{Ag}$, and Cu)": P. Jerabek, B. von der Esch, H. Schmidbaur, P. Schwerdtfeger, *Inorg. Chem.*, **56**, 14624–14631, 2017.
24. "The Oxygen-Rich Beryllium Oxides BeO_4 and BeO_6 ": Q. Zhang, P. Jerabek, M. Chen, M. Zhou, G. Frenking, *Angew. Chem. Int. Ed.*, **55**, 10863–10867, 2016.
23. "Understanding the F 1s NEXAFS Dichroism in Fluorinated Organic Semiconductors": M. Klues, P. Jerabek, T. Breuer, M. Oehzelt, K. Hermann, R. Berger, G. Witte, *J. Phys. Chem. C*, **120**, 12693–12705, 2016.
22. "Zn...Zn interactions at nickel and palladium centers": K. Freitag, M. Molon, P. Jerabek, K. Dilchert, C. Rösler, R. W. Seidel, C. Gemel, G. Frenking, R. A. Fischer, *Chem. Sci.*, **7**, 6413–6421, 2016.
21. "The σ -Aromatic Clusters $[\text{Zn}_3]^+$ and $[\text{Zn}_2\text{Cu}]$: Embryonic Brass": K. Freitag, C. Gemel, P. Jerabek, M. I. Opperl, R. W. Seidel, G. Frenking, H. Banh, K. Dilchert, R. A. Fischer, *Angew. Chem. Int. Ed.*, **54**, 4370–4374, 2015.
20. "Stepwise Synthesis of Siloxane-Substituted Oligoarsanes and Structural Investigation of Alkaline Earth Metal Derivatives": C. Clobes, P. Jerabek, I. Nußbruch, G. Frenking, C. von Hänisch, *Eur. J. Inorg. Chem.*, 3264–3273, 2015.
19. "Isolation of Neutral Mofalsenuclear Copper Complexes Stabilized by Two Cyclic (Alkyl)(amino)carbenes": D. S. Weinberger, N. Amin SK, K. C. Mondal, M. Melaimi, G. Bertrand, A. C. Stückl, H. W. Roesky, B. Dittrich, S. Demeshko, B. Schwederski, W. Kaim, P. Jerabek, G. Frenking, *J. Am. Chem. Soc.*, **136**, 6235–6238, 2014.
18. "A Novel Concept for the Synthesis of Multiply Doped Gold Clusters $[(\text{M}@\text{Au}_n\text{M}'_m)\text{Lk}]_q^+$ ": A. Puls, P. Jerabek, W. Kurashige, M. Molon, T. Bollermann, M. Winter, C. Gemel, Y. Negishi, G. Frenking, R. A. Fischer, *Angew. Chem. Int. Ed.*, **53**, 4327–4331, 2014.
17. "Experimental Charge Density Study of a Silylone": B. Niepötter, R. Herbst-Irmer, D. Kratzert, P. P. Samuel, K. C. Mondal, H. W. Roesky, P. Jerabek, G. Frenking, D. Stalke, *Angew. Chem. Int. Ed.*, **53**, 2766–2770, 2014.

16. "Hume-Rothery Phase-Inspired Metal-Rich Molecules: Cluster Expansion of $[\text{Ni}(\text{ZnMe})_6(\text{ZnCp}^*)_2]$ by Face Capping with $\text{Ni}(0)(\eta(6)\text{-toluene})$ and $\text{Ni}(\text{I})(\eta(5)\text{-Cp}^*)$ ": M. Molon, C. Gemel, P. Jerabek, L. Trombach, G. Frenking, R. A. Fischer, *Inorg. Chem.*, **53**, 10403–10411, 2014.
15. "Coinage Metals Binding as Main Group Elements: Structure and Bonding of the Carbene Complexes $[\text{TM}(\text{cAAC})_2]$ and $[\text{TM}(\text{cAAC})_2]^+$ (TM = Cu, Ag, Au)": P. Jerabek, H. W. Roesky, G. Bertrand, G. Frenking, *J. Am. Chem. Soc.*, **136**, 17123–17135, 2014.
14. "Comparative bonding analysis of N_2 and P_2 versus tetrahedral N_4 and P_4 ": P. Jerabek, G. Frenking, *Theor. Chem. Acc.*, **133**, 1447–1456, 2014.
13. "Bonding analysis of telluroketones $\text{H}_2\text{A}=\text{Te}$ (A = C, Si, Ge)": N. B. Jaufeerally, P. Ramasami, P. Jerabek, G. Frenking, *J. Mol. Mod.*, **20**, 2433–2440, 2014.
12. "Dizinc Cation $[\text{Zn}_2]^+$ Trapped In A Homoleptic Metalloid Coordination Environment Stabilized by Dispersion Forces: $[\text{Zn}_2(\text{GaCp}^*)_6][\text{BAr}_4\text{F}]^2$ ": K. Freitag, H. Banh, C. Gemel, P. Jerabek, G. Frenking, R. A. Fischer, *Inorg. Chem.*, **54**, 352–358, 2014.
11. "Isolation of Neutral Mofalse- and Dinuclear Gold Complexes of Cyclic (Alkyl)(amino)carbenes": D. S. Weinberger, M. Melaimi, C. E. Moore, A. L. Rheingold, G. Frenking, P. Jerabek, G. Bertrand, *Angew. Chem. Int. Ed.*, **52**, 8964–8967, 2013.
10. "The Orgafalsezinc Rich Compounds $[\text{Cp}^*\text{M}(\text{ZnR})_5]$ (M = Fe, Ru; R = Cp^* , Me, Cl, Br)": M. Molon, C. Gemel, R. W. Seidel, P. Jerabek, G. Frenking, R. A. Fischer, *Inorg. Chem.*, **52**, 7152–7160, 2013.
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